The University of Nottingham Ningbo China

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER, 2023–2024

COMPUTER FUNDAMENTALS

Time allowed 60 Minutes

Candidates may complete the front covers of their answer books and sign their desk cards but must NOT write anything else until the start of the examination period is announced.

Answer all questions. The total mark is 50.

Only silent, self contained calculators with a Single-Line Display or Dual-Line Display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

Question 1 (25 marks)

(a) Convert the following 8-bit two's complement binary into decimal. $a = 1110\ 1011_2$ $b = 1110\ 0011_2$ $c = 0111\ 0001_2$ (3 marks)

(b) Convert the following decimal values to 8-bit two's complement binary. $a = -117$ $b = -12$ $c = 49$ (3 marks)

(c) Simplify the following Boolean Expression.

(i) $B + (D(AD + \overline{CB}) + AD)C$

(ii) $BA + (\overline{CB} + \overline{B}C)\overline{C}$

(iii) $CD + (DA + \overline{C}C)B$ (6 marks)

(d) Draw the sequential logic circuit diagram for 1-bit register using **MUX** and **DFF**. (5 marks)

(e) Program counter (PC) emits the address of the next instruction. Please briefly describe the **four** key working steps of the PC. (8 marks)

END OF QUESTION 1

Question 2 (25 marks)

(a) This question is based on the symbolic assembly code below:

```
   @0
   M=1
   @2
   D=A
   @1
   M=D
   @2
   M=0
   @0
   D=M
   @END
   D;JEQ
   @1
   D=M
   @END
   D;JEQ
   @0
   D=M
   @3
   M=D
(LOOP)
   @1
   D=M
   @2
   M=D+M
   @3
   M=M-1
   D=M
   @LOOP
   D;JGT
(END)
   @END
   0;JMP
```

(i) Derive the value of register **RAM[2]** after the execution of this piece of code. (4 marks)

(ii) Convert the first two lines of the symbolic assembly code in (a),

```
@0
M=1
```

to binary machine code. You may refer to APPENDIX 1 and 2 for this conversion. (2 marks)

(b) Please offer definitions for the following terms:

(i) immediate addressing (1 mark)

(ii) direct addressing (1 mark)

(iii) indirect addressing (1 mark)

(c) Draw the final stack status after running VM code below: (4 marks)

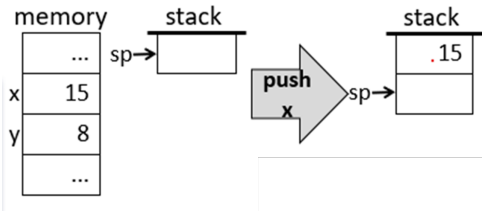
```
push constant 7
push constant 7
eq
push constant 2
push constant 1
lt
push constant 7
push constant 6
gt
push constant 57
push constant 31
push constant 53
add
push constant 112
sub
neg
and
push constant 82
or
not
```

(d) Write the pseudo VM code for the following expression: $\boldsymbol{d = (2 - x) + (y + 9)}$ (2 marks)

(e) Show the stack operations and the final memory status for the following operations: $\boldsymbol{(x > 7) and (y > 7)}$
The initial memory status and the first stack operation are shown below. (5 marks)

(f) Translate the VM command **"push local 5"** into **Hack symbolic assembly code**. (5 marks)

END OF QUESTION 2



APPENDIX 1 A-instruction specification

Symbolic syntax:

@value

Binary syntax:

0value

APPENDIX 2 C-instruction specification

Symbolic syntax: *dest* = *comp* ; *jump*

Binary syntax: 1 1 1 a c1 c2 c3 c4 c5 c6 d1 d2 d3 j1 j2 j3

opcode | not used | *comp* bits | *dest* bits | *jump* bits

comp		c1	c2	c3	c4	c5	c6
0		1	0	1	0	1	0
1		1	1	1	1	1	1
-1		1	1	1	0	1	0
D		0	0	1	1	0	0
A	M	1	1	0	0	0	0
!D		0	0	1	1	0	1
!A	!M	1	1	0	0	0	1
-D		0	0	1	1	1	1
-A	-M	1	1	0	0	1	1
D+1		0	1	1	1	1	1
A+1	M+1	1	1	0	1	1	1
D-1		0	0	1	1	1	0
A-1	M-1	1	1	0	0	1	0
D+A	D+M	0	0	0	0	1	0
D-A	D-M	0	1	0	0	1	1
A-D	M-D	0	0	0	1	1	1
D&A	D&M	0	0	0	0	0	0
D|A	D|M	0	1	0	1	0	1
a==0	a==1						

dest	d1	d2	d3	effect: the value is stored in:
null	0	0	0	The value is not stored
M	0	0	1	RAM[A]
D	0	1	0	D register
MD	0	1	1	RAM[A] and D register
A	1	0	0	A register
AM	1	0	1	A register and RAM[A]
AD	1	1	0	A register and D register
AMD	1	1	1	A register, RAM[A], and D register

jump	j1	j2	j3	effect:
null	0	0	0	no jump
JGT	0	0	1	if out > 0 jump
JEQ	0	1	0	if out = 0 jump
JGE	0	1	1	if out ≥ 0 jump
JLT	1	0	0	if out < 0 jump
JNE	1	0	1	if out ≠ 0 jump
JLE	1	1	0	if out ≤ 0 jump
JMP	1	1	1	Unconditional jump